DRAFT

Utah Unây – Robotics and TEK

WHOI David Center, 6/13/26-6/14/26, 9-3:30

Utah Unây Leadership:

Leslie Jonas, WHOI Tribal Liaison, ljonas@whoi.edu

Casey Thornbrugh, Just Collective Co-Director, casey@justcollective.net

Grace Simpkins, WHOI Sea Grant Marine Education Specialist, gsimpkins@whoi.edu

Ben Weiss, Master's Student at MIT, benw2@mit.edu

DAY 1

	Introduction to Oceanographic Sensing and Instrumentation
Time: 9:00-9:30 **Leaders:** Leslie, Grace, Ben	General Introductions ▪ Smudge ▪ Inside: introduce everyone - names, where you are from, fun fact ▪ General overview of WHOI (Ben ~1 minute) ▪ General overview of weekend (Grace)
Time: 9:30-9:45 **Leader:** Clayton	Safety Briefing
Time: 9:45-10:15 **Leader:** Ben	Walkthrough of David Center for Innovation
	Traditional Ecological Knowledge
Time: 10:15-10:45 **Leader:** Leslie	Two-Eyed Seeing: Looking at the world through two lenses
Time: 10:45-11:00	Ice Breaker Name, favorite thing to do outside, and what is one thing in the natural world that you wish you knew more about
Time: 11 - 11:45 **Leader:** Jordan Kennedy, Darius Coombs, Paula	Two-Eyed Seeing (Hands-on Wampum Activities) ● 3 stations that groups cycle through
Time: 11:45 – 12:45	LUNCH (Culture Keepers)
	Microcontrollers and Sensing

Time: 12:45 – 1:45 **Leader:** Ben	• What is a microcontroller? • Download Arduino IDE • Download example files from Github • Running Your First Script: Blink an LED on a Feather CODE_1
Time: 1:45 – 2:15 **Leader:** Ben	Sensing Light CODE_2
Time: 2:15 – 2:30 **Leader:** Ben	Sensing Temperature and Humidity CODE_3
Time: 2:30-2:45 **Leader:** Ben	Data Logging: Adding an SD Card CODE_4
Time: 2:45-3 **Leader:** Ben	Visualizing Data: Graphing
Time: 3-3:15 **Leader:** Ben	Make it deployable! • Walk down to the beach, put your sensor in the water and test
Time: 3:15-3:30	Wrap up • If we don't get to ice breaker in AM, discuss here

DAY 2

	Making Your System Interactive
Time: 9-9:20 **Leader:** Ben	Add screen • Run example script to show splash screen
	Asking good questions (think, pair, share) cont…
Time: 9:20 – 9:40 **Leader:** Grace	We give each pair of students a different prompt and they must pick the sensors they would use on their device to answer the questions • List the sensors • Where, when, how long would you deploy it for? • What data would you collect? • How would it answer your questions?
	Experimental Design and Deployment
Time: 9:40-10 **Leader:** Ben	• What scientific question are you interested in answering for yourself? ○ Think back to what things in nature you were interested in exploring yesterday

	● What sensor do you need to answer that question? ● Think, pair, share
Time: 10 - 10:30 **Leader:** Ben	Deploy Around DCOI- decide if case should be open or closed This could be… ● In a test tank ● In the woods ● In DCOI
	Traditional Ecological Knowledge
Time: 10:30 - 11:30 **Leader:** Mother Bear	Mother Bear Activity
Time: 11:30-12:20	LUNCH (Culture Keepers)
	Data Analysis and Graphing
Time: 12:20-12:40 **Leader:** Ben	● Use the second script to graph new data stream in addition to the three that were already being plotted (temp, humidity, light intensity)
Time: 12:40-1:00 **Leader:** Ben	● Discussion on interpreting the data and drawing conclusions (give multiple examples) ● Students interpret their own data and share with a partner for feedback
	Presentations 101 and Telling the Story of the Data
Time: 1:00-2 **Leader:** Grace	Why is it important for us to present/represent our work (as scientist/engineers) ● Know your audience — different formats for different audiences ● Share out Circle Prep o Intro yourself o Pose your question that you wanted to solve o Describe your system (use props) o Describe your results o Future directions - what would you want to use your sensor system for next? (You may want to think about your science fair) ● Have everyone give a practice presentation
	Share Out Circle
Time: 2-3:30	Share out circle presentations ● Invite friends and family ● Food!! ● Have everyone introduce themselves o How you are related to other people at the circle

	o Some other fun fact… • Present students with certificates • Take group photo